

Laptop Price & Specification Analysis

Power BI Dashboard Report | Indian Market

Python Analysis + Power BI Dashboard

Domain: E-commerce / Consumer Electronics

Dataset: laptop_price.csv (1,370 rows x 13 columns)

Analysis period stated in project: 2024-2025

1. Project Overview

This project analyses laptop products, specifications and prices to understand the Indian laptop market. The work starts with dataset understanding and cleaning in Python, continues with feature transformation and exploratory analysis, and finishes with a Power BI dashboard that presents the findings in a simple, business-friendly format.

Objective

To understand how laptop brands, specifications and laptop types are related to price, and to present the important findings through an interactive Power BI dashboard.

Project Flow

- Dataset selection and understanding
- Data cleaning and preparation
- Feature transformation
- Exploratory data analysis
- Business insights and recommendations
- Power BI dashboard creation

Project Summary

Item	Detail
Project type	Data Analysis and Business Intelligence
Domain	E-commerce / Consumer Electronics
Dataset	laptop_price.csv (source: GitHub)
Size	1,370 rows x 13 columns
Tools	Python, Power BI
Final output	Cleaned dataset + analytical findings + interactive dashboard

2. Dataset & Data Preparation

Main Features

Feature	Meaning
laptop_ID	Unique laptop record identifier
Company	Laptop brand / company
Product	Laptop model name
Type	Category such as Notebook, Gaming, Ultrabook, 2 in 1 Convertible
Inches	Screen size
Cpu / Gpu	Processor and graphics processor

Ram / Memory	RAM capacity and storage configuration
OpSys	Operating system
Weight	Laptop weight
Price	Laptop price

Cleaning Performed

Step	What was done
Missing values	Missing-value check performed; zero missing values recorded
Duplicate records	Duplicate check performed; 0 duplicates
Weight / RAM	Removed "kg" and "GB" units and converted both fields to numeric
Column names	Simplified names, e.g. TypeName to Type, Price_euros to price
Unwanted columns	Product and ScreenResolution removed for later analysis
Price outliers	IQR-based treatment used to identify and remove unusual price values
Skewness	Price skewness checked and log transformation applied to reduce its effect

Feature Transformation

Feature	Purpose
RAM_GB, Screen_size, Weight_kg	Numeric fields for calculations and comparisons
Ram_category	Groups RAM into LOW, MEDIUM, HIGH, VERY HIGH and EXTREME HIGH
Weight_per_Inch	Compares laptop weight with screen size (portability vs. design)
discounted_price / final_price	Price after the 10% discount scenario used in the project
price_inr	Project-defined Indian-price field
price_log	Log-transformed price to reduce skewness

The cleaned and transformed dataset was exported as the project's cleaned CSV and used as the source for the Power BI dashboard.

3. Exploratory Data Analysis

Level	Analysis	Main finding
Univariate	Company count, price histogram, weight box plot	Some brands have much higher product representation; products concentrate in lower-to-middle price ranges

Bivariate	Type vs price, type vs screen size, RAM vs screen size	Laptop types occupy different price ranges; RAM and screen size show a weak positive relationship (about 0.19)
Multivariate	RAM + screen + weight + OS; RAM category + company; weight + type + screen	Higher RAM does not automatically mean higher weight; larger/heavier designs are linked to Gaming and Workstation, smaller designs to Ultrabook/Netbook

4. Dashboard Overview

The Power BI report contains three pages. Each page answers a different business question, and the pages share one consistent navy-and-teal theme.

Page	Title	Focus
1	Laptop Price & Marketing Analysis (India Market)	Brand performance, revenue, OS share and price segments
2	Comprehensive Laptop Market & Price Analysis	Brand, OS and RAM-category analysis; global distribution
3	Laptop Specification & Value Analysis	Type share, storage, RAM, weight and screen size

Page 1: Laptop Price & Marketing Analysis

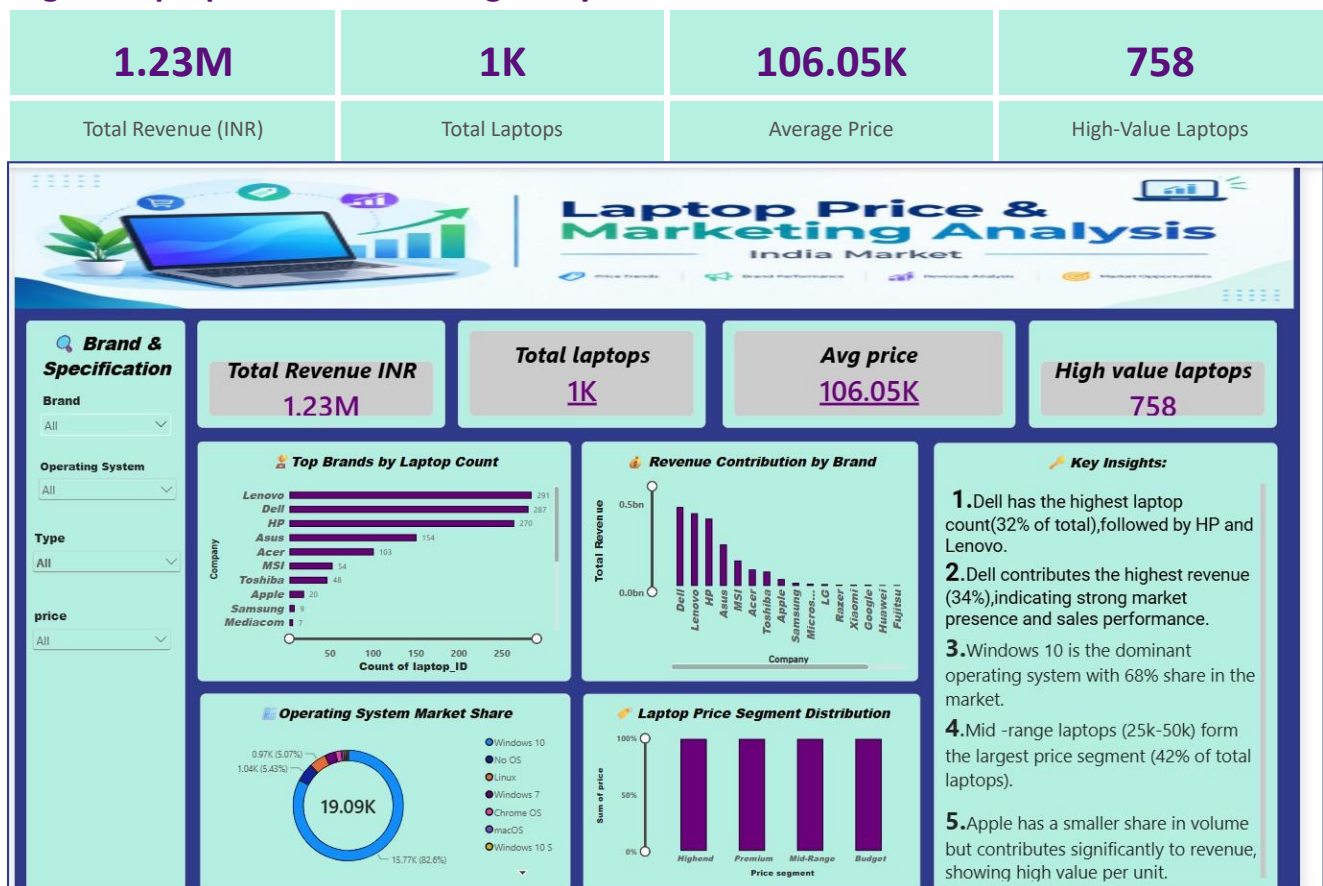


Figure 1: Marketing analysis page with slicers (Brand, Operating System, Type, Price)

Page 1: What the visuals show

- **Top Brands by Laptop Count:** Lenovo (291), Dell (287) and HP (270) lead the catalogue, followed by Asus (154) and Acer (103).

- **Revenue Contribution by Brand:** Dell, Lenovo and HP generate the largest revenue; Apple contributes strongly despite a small unit count (20 laptops).
- **Operating System Market Share:** Windows 10 holds about 82.6% of the market, far ahead of No OS, Linux and others.
- **Price Segment Distribution:** laptops are grouped into Budget, Mid-Range, Premium and High-end segments.
- **Slicers:** Brand, Operating System, Type and Price allow interactive filtering of every visual.

Page 2: Comprehensive Laptop Market & Price Analysis

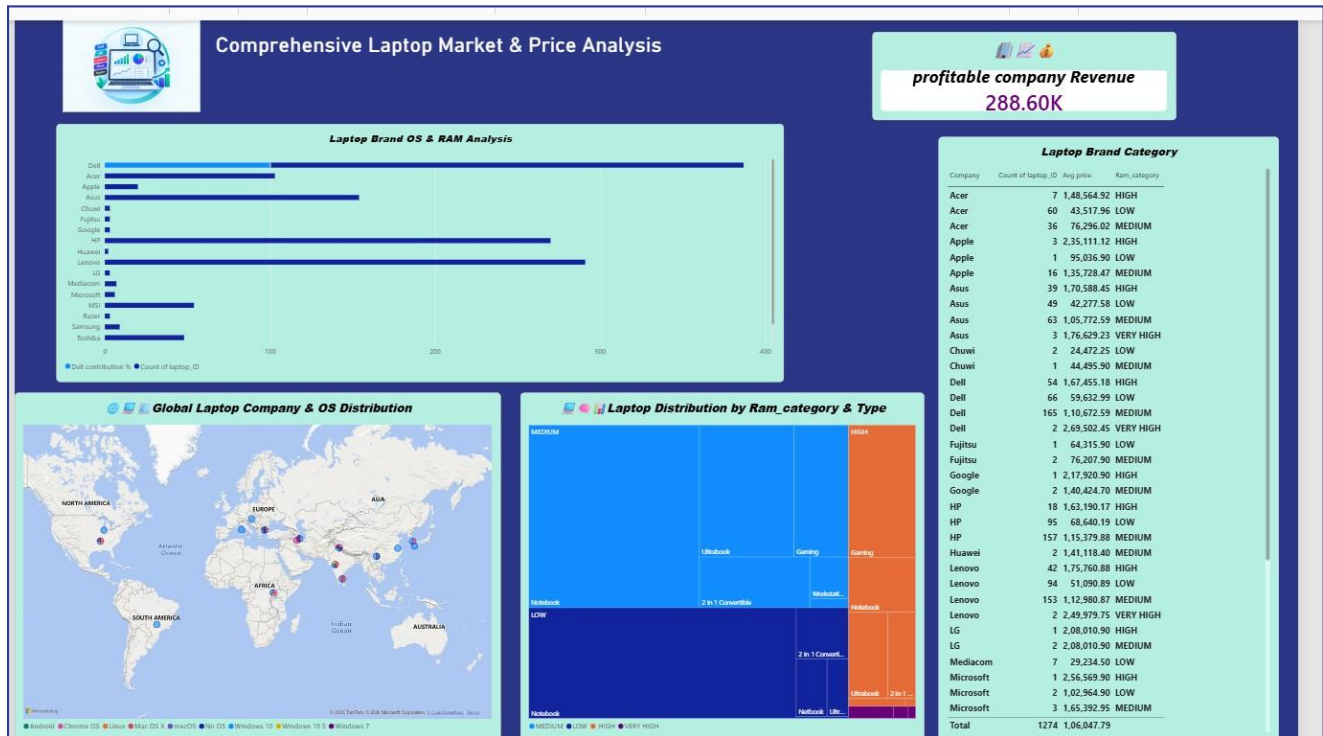


Figure 2: Brand, OS and RAM-category analysis with global distribution map

- **Brand OS & RAM Analysis:** compares laptop counts across brands, with Dell contribution % highlighted.
- **Laptop Brand Category table:** average price by brand and RAM category (LOW, MEDIUM, HIGH, VERY HIGH); price rises steadily with RAM category for every major brand.
- **Distribution by RAM category & Type:** a treemap showing Notebook dominates both MEDIUM and LOW RAM, while Gaming and Ultrabook appear more in HIGH RAM.
- **Profitable company revenue card:** 288.60K for the top company.

Page 3: Laptop Specification & Value Analysis

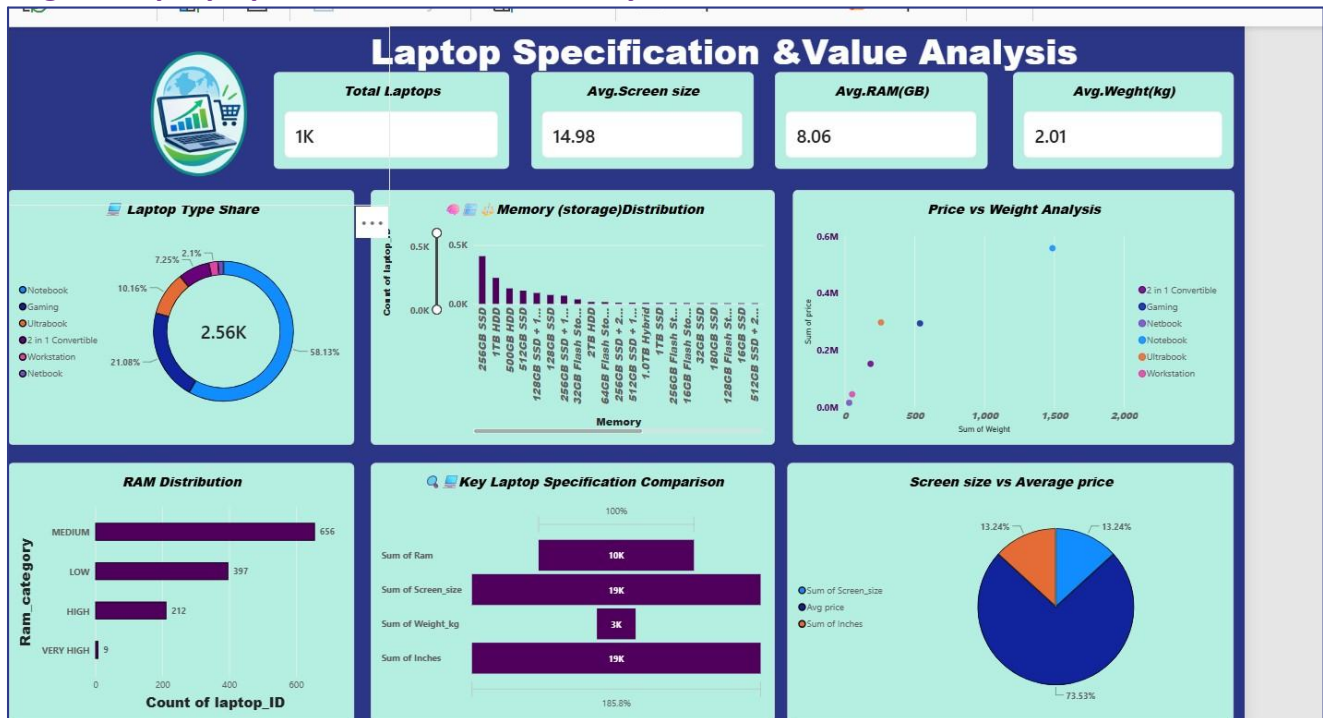


Figure 3: Specification and value analysis page

- **Laptop Type Share:** Notebook is the largest category (58.13%), followed by Gaming (21.08%) and Ultrabook (10.16%).
- **Memory (storage) Distribution:** 256GB SSD is the most common configuration, followed by 1TB HDD and 500GB HDD.
- **RAM Distribution:** MEDIUM 656, LOW 397, HIGH 212, VERY HIGH 9 laptops.
- **Price vs Weight and Screen Size analysis:** compares price against physical specifications across laptop types.

5. Key Findings & Recommendations

Key Findings

- **Price drivers:** RAM and CPU are strong price-related factors; laptops with higher RAM and stronger processors have higher average prices.
- **Brands:** Dell, Lenovo and HP dominate volume and revenue; Apple is premium-focused, with high value per unit.
- **Laptop type:** Ultrabook and Gaming categories are priced above Notebook, making type an important segmentation variable.
- **Physical specs:** RAM and weight show no strong direct relationship; screen size and weight are positively related; most weights sit around 1.5 to 2.5 kg.
- **Operating system:** Windows 10 is the dominant OS in the dashboard (about 82.6%).

Business Recommendations

- Use brand and laptop type together for inventory and pricing decisions.
- Target lightweight laptops toward students and office users.
- Target higher-priced gaming / high-performance laptops toward performance-focused customers.
- Use RAM and CPU as the main specification dimensions when creating price segments.